A Paley–II analogue of the Scarpis–Đoković construction for prime powers $q \equiv 1 \pmod 4$

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Abstract

A Hadamard matrix of order m is a $\{1,-1\}$-matrix H with $HH^T = mI_m$. Scarpis (1898) showed that for a prime $p \equiv 3 \pmod 4$ a Hadamard matrix of order $p+1$ can be lifted to one of order $p(p+1)$, and Đoković extended the lift to all prime powers $q \equiv 3 \pmod 4$, leaving open the analogous procedure for $q \equiv 1 \pmod 4$. We give an explicit, uniform construction of a Hadamard matrix of order $2q(q+1)$ for every prime power $q \equiv 1 \pmod 4$, built from the quadratic character on $\mathbb{F}_q$ and the 2×2 Paley–II replacement blocks; since the Paley–II seed has order $2(q+1)$, this is the natural Paley–II analogue of the lift. The orders so produced are not new existence results. Every order $2q(q+1)$ whose odd part is at most 3000 is already recorded in the standard tables, and the smallest order lying beyond that audited range, $q = 81$ (order $13284 = 4\cdot 41\cdot 81$), is a classical Turyn product of T-matrices of order 41 with Williamson-type matrices of order $81 = 9^2$. We make the coverage explicit, exhibit the witnessing constructions, identify the smallest unaudited order, and tabulate those orders up to $q = 2917$ for which no elementary witness is presently evident, emphasising that these are *unaudited* rather than open.

1 Introduction

For a prime power q let $\mathbb{F} = \mathbb{F}_q$ and let $\chi\colon \mathbb{F} \to \{0,\pm 1\}$ be the quadratic character, $\chi(0) = 0$, $\chi(x) = 1$ for nonzero squares and -1 otherwise. Paley's two constructions split on the residue of q: for $q \equiv 3 \pmod 4$ Paley I gives a Hadamard matrix of order $q+1$, and for $q \equiv 1 \pmod 4$ Paley II gives one of order $2(q+1)$ [2].

Scarpis [3] showed that for a prime $p \equiv 3 \pmod 4$ a Hadamard matrix of order $p+1$ lifts to one of order $p(p+1)$. Đoković [7] replaced the prime by an arbitrary prime power: if $q \equiv 3 \pmod 4$ is a prime power there is a Hadamard matrix of order $q(q+1)$; his note closes with the open problem of an analogous procedure for $q \equiv 1 \pmod 4$.

Because the Paley–II seed has order $2(q+1)$, the natural Scarpis-type target in the $q \equiv 1 \pmod 4$ case is

$$q \cdot 2(q+1) = 2q(q+1).$$

We construct such a matrix directly from χ. Throughout, $q \equiv 1 \pmod 4$ is a prime power, so $\chi(-1) = 1$; this is the only place the residue class is used, and it is exactly what fails for $q \equiv 3 \pmod 4$.

2 The Paley–II blocks

Put

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and let $\psi\colon \{0,\pm1\} \to \{\pm1\}^{2\times 2}$ be $\psi(0) = Z$, $\psi(1) = P$, $\psi(-1) = -P$; these are the Paley–II replacement blocks, and $ZR = -P$. Write $\mathcal{R} = J_q \otimes I_2$, where J_q is the $q \times q$ all-ones matrix. We use repeatedly

$$\sum_{u\in\mathbb{F}} \psi(\chi(u)) = Z, \tag{2.1}$$

which holds because the nonzero squares and non-squares are equinumerous.

3 The construction

For $t \in \mathbb{F}$ let K_t be the $2q \times 2q$ matrix with rows and columns indexed by pairs (x, a), $x \in \mathbb{F}$, $a \in \{0, 1\}$, whose 2×2 block in position (x, y) is

$$K_t[x, y] = \psi\big(\chi(y - x - t)\big).$$

Lemma 1. *For every $t \in \mathbb{F}$, $K_tK_t^T = 2(q + 1)I_{2q} - 2\mathcal{R}$.*

Proof. The (x, x') block of $K_tK_t^T$, with $d = x' - x$ and $z = y - x - t$, equals $\sum_z \psi(\chi(z))\psi(\chi(z-d))^T$. For $d = 0$ each summand is $2I_2$, giving $2qI_2$. For $d \neq 0$ the terms $z = 0$ and $z = d$ carry opposite skew parts and cancel, using $\chi(-d) = \chi(d)$ (here $\chi(-1) = 1$); for $z \neq 0, d$, $\psi(\chi(z))\psi(\chi(z - d))^T = 2\chi(z)\chi(z - d)I_2$, and $\sum_z \chi(z(z - d)) = -1$ yields $-2I_2$. These are the diagonal and off-diagonal blocks of $2(q + 1)I - 2\mathcal{R}$. □

Lemma 2. *For $r, s \in \mathbb{F}$,*

$$\sum_{a\in\mathbb{F}} K_{ar}K_{as}^T = \begin{cases} 2q(q + 1)I_{2q} - 2q\mathcal{R}, & r = s, \\ 2\mathcal{R}, & r \neq s. \end{cases}$$

Proof. The case $r = s$ is q copies of Lemma 1. For $r \neq s$, in the (x, x') block put $z = y - x - ar$; the second argument becomes $z - (x' - x) + a(r - s)$, which ranges over all of $\mathbb{F}$ independently of z as a does. By (2.1) each block is $ZZ^T = 2I_2$, i.e. the corresponding block of $2\mathcal{R}$. □

For $r \in \mathbb{F}$ let B_r be the $2q \times 2q$ matrix whose row (x, a) equals row (r, a) of K_0 for every x; thus B_r has only two distinct rows. From Lemma 1, $B_rB_s^T = 2q\mathcal{R}$ if $r = s$ and $-2\mathcal{R}$ if $r \neq s$.

Form the $2q \times 2q(q + 1)$ block rows $M_r = \big[\, B_r \mid K_{ar}\ (a \in \mathbb{F}) \,\big]$.

Lemma 3. *$M_rM_s^T = 2q(q + 1)I_{2q}$ if $r = s$, and 0 if $r \neq s$.*

Proof. $M_rM_s^T = B_rB_s^T + \sum_a K_{ar}K_{as}^T$. For $r = s$ the $2q\mathcal{R}$ of the border cancels the $-2q\mathcal{R}$ of Lemma 2; for $r \neq s$ the $-2\mathcal{R}$ cancels the $2\mathcal{R}$. □

Let C be the Paley conference matrix indexed by $\{\infty\} \cup \mathbb{F}$, namely $C_{u,u} = 0$, $C_{\infty,x} = C_{x,\infty} = 1$, $C_{x,y} = \chi(y - x)$, and let S be the order-$2(q + 1)$ Paley–II Hadamard matrix obtained by replacing $0, 1, -1$ by $Z, P, -P$. Delete the two rows of S indexed by ∞; in each remaining row repeat the ∞-column pair q times unchanged, and replace each finite column pair by its product with R, repeated q times. This is a $2q \times 2q(q + 1)$ matrix M_∞.

Lemma 4. *$M_\infty M_\infty^T = 2q(q + 1)I_{2q}$ and $M_\infty M_r^T = 0$ for all $r \in \mathbb{F}$.*

Proof. The first identity holds because S is Hadamard, each column pair is repeated q times, and right multiplication by R preserves row inner products. For the cross identity, summing a cap row against a row of M_r and using (2.1) on every $2q$-block leaves $\big(P_\alpha + (\sum_a S[e, a]_\alpha)R\big) \cdot Z_\beta$; the inner sum of finite entries of a row of S is Z_α, and $P + ZR = P - P = 0$. □

Theorem 1. *Let $q \equiv 1 \pmod 4$ be a prime power. Then*

$$H = \begin{bmatrix} M_\infty \\ M_r \ (r \in \mathbb{F}) \end{bmatrix}$$

is a Hadamard matrix of order $2q(q+1)$.

Proof. H is $\{1, -1\}$ and square of order $2q + q \cdot 2q = 2q(q+1)$. Lemmas 3 and 4 give $HH^T = 2q(q+1)I$. □

Corollary 1. *For every prime power $q \equiv 1 \pmod 4$ there is a Hadamard matrix of order $2q(q+1)$; the smallest non-prime instance is $q = 9$, of order 180.*

The construction has been verified by full Gram computation for all $q \leq 49$ (orders $60, \ldots, 4900$, including the extension fields $q = 9, 25, 49$) and by streamed row-orthogonality checks for $q = 81$ and $q = 125$ (orders 13284 and 31500).

4 Remarks on novelty

Theorem 1 supplies, for the $q \equiv 1 \pmod 4$ case, the uniform family that Đoković [7] asked for. It does not, however, produce Hadamard matrices of previously unconstructible orders, and we record why.

Since $q \equiv 1 \pmod 4$, the 2-adic valuation of the order is exactly 2:

$$2q(q+1) = 4 \cdot q \cdot \tfrac{q+1}{2}, \qquad q \text{ and } \tfrac{q+1}{2} \text{ both odd.} \tag{4.1}$$

Thus every order is $4 \times (\text{odd})$, so it is never a Kronecker product of two smaller Hadamard matrices, and its odd part $q(q+1)/2$ is always composite. Two consequences follow. First, the family never meets the dominant list of open Hadamard orders, which are $4 \times$ prime. Second, (4.1) is itself a Turyn factorisation: with T-matrices of order t and Williamson-type matrices of order w one obtains a Hadamard matrix of order $4tw$ [5, 6], and here $\{t, w\} = \{q, (q+1)/2\}$. The coverage analysis of Appendix A makes both points quantitative: all orders with odd part ≤ 3000 are recorded in the standard tables, the first order beyond that range is itself a Turyn product, and the orders for which no elementary witness is currently evident are listed but are unaudited rather than open.

A Coverage analysis

Write $N = 2q(q+1)$ and $m = q(q+1)/2$ for its odd part, so $N = 4m$. We assess coverage of N by prior constructions on three levels: the audited tables (§A.1), the first unaudited order (§A.2), and an elementary-witness test that isolates candidate orders (§A.3).

A.1 Orders with odd part at most 3000

The surveys of Seberry and Yamada [6, 9] and the database of Cati and Pasechnik [10] record, for every odd $m \leq 3000$, the least t for which a Hadamard matrix of order $2^t m$ is known. The default is $t = 2$ (so $N = 4m$ is known), and the only odd $m \leq 3000$ with $t > 2$ are the primes $167, 179, 223$ (orders $668, 716, 892$, still open). Since $m = q(q+1)/2$ is composite it is never one of these; hence *every N with $m \leq 3000$ is a known Hadamard order.* This is the range $q \leq 73$, tabulated below with an explicit elementary witness where one exists (Paley I/II, or a Turyn product $4tw$).

q	$N = 2q(q+1)$	odd part m	witnessing construction
5	60	$3 \cdot 5$	Paley II, $p = 29$
9	180	$3^2 \cdot 5$	Paley II, $p = 89$
13	364	$7 \cdot 13$	Paley II, $p = 181$
17	612	$3^2 \cdot 17$	Turyn $4 \cdot 9 \cdot 17$
25	1300	$5^2 \cdot 13$	Turyn $4 \cdot 13 \cdot 25$
29	1740	$3 \cdot 5 \cdot 29$	Turyn $4 \cdot 15 \cdot 29$
37	2812	$19 \cdot 37$	Turyn $4 \cdot 19 \cdot 37$
41	3444	$3 \cdot 7 \cdot 41$	Paley II, $p = 1721$
49	4900	$5^2 \cdot 7^2$	Turyn $4 \cdot 25 \cdot 49$
53	5724	$3^3 \cdot 53$	Paley II, $p = 2861$
61	7564	$31 \cdot 61$	Turyn $4 \cdot 31 \cdot 61$
73	10804	$37 \cdot 73$	tables; no elementary witness (see §A.3)

The entry $q = 73$ is instructive: its odd part $2701 = 37 \cdot 73$ carries the same obstruction we use below (the prime 73), yet $N = 10804$ is a known Hadamard order because $m \leq 3000$ is audited. Obstruction in the elementary test is therefore a property of that test, not of Hadamard existence.

A.2 The first unaudited order: $q = 81$

The smallest order of the family whose odd part exceeds 3000, and hence lies beyond the audited range, is

$$q = 81 = 3^4, \qquad N = 13284, \qquad m = 3321 = 3^4 \cdot 41 > 3000.$$

It is nevertheless a known Hadamard order. By (4.1),

$$13284 = 4 \cdot 41 \cdot 81 = 4tw, \qquad t = 41,\ w = 81,$$

and both ingredients are classical: T-matrices of order 41 come from the base sequences $BS(21, 20)$ [5, 8], and Williamson-type matrices of order $81 = 9^2$ come from Turyn's 9-power family [5, 6]. The Turyn array on these [6, 9] yields a Hadamard matrix of order $4 \cdot 41 \cdot 81 = 13284$. Thus $q = 81$ is the smallest unaudited order, and it is covered.

A.3 An elementary-witness test and the candidate orders

Beyond the audited range we have no table to appeal to, so we apply a deliberately conservative *elementary-witness* test based on (4.1) and Turyn's array. Call an odd prime p *T-eligible* if $p \leq 59$ (so that T-matrices of order p follow from base sequences $BS(m{+}1, m)$, $m \leq 29$) and *W-eligible* if $p \leq 33$ or $2p - 1$ is a prime power (Williamson-type matrices of order p, the latter by Turyn's $(P{+}1)/2$ family with $P = 2p - 1 \equiv 1 \pmod 4$); both classes are multiplicatively closed. An order $N = 4m$ passes the test, and is then a Hadamard order, if N is a Paley I/II order or if every prime factor of m is T- or W-eligible. A prime factor p of m is *obstructing* if $p > 59$ and $2p - 1$ is not a prime power.

A passing verdict is a genuine construction; a failing verdict (*no elementary witness*) is a statement about this test only, not a claim that N is open. Over the 232 orders with $q \leq 3000$, the test passes 131 (audited, Paley, or Turyn) and fails 101. The smallest failure is

$$q = 109, \qquad N = 23980, \qquad m = 5 \cdot 11 \cdot 109,$$

obstructed by 109 ($2 \cdot 109 - 1 = 217 = 7 \cdot 31$ is not a prime power). This $N = 23980$ is the smallest order of the family for which we exhibit neither an audited entry nor an elementary witness; we do not assert it is open, and indeed Williamson-type matrices of order 109, were they tabulated, would settle it at once (compare $q = 73$).

Table 1 lists all such candidate orders up to $q = 2917$. The previously discussed $q = 2689$ and $q = 2917$ appear at the end; both are obstructed (2689 and 2917 are primes with $2 \cdot 2689 - 1 = 5377 = 19 \cdot 283$ and $2 \cdot 2917 - 1 = 5833 = 19 \cdot 307$), and $q = 2917$ has the extreme form $m = 1459 \cdot 2917$, a product of two obstructing primes. They are not distinguished among the candidates except by size.

Table 1: Orders $N = 2q(q+1)$, q a prime power $\equiv 1 \pmod 4$, with no elementary witness (§A.3), $109 \le q \le 2917$. "obstr." lists the obstructing prime factors of the odd part $m = q(q+1)/2$.

q	N	odd part m	obstr.
109	23980	$5 \cdot 11 \cdot 109$	109
173	60204	$3 \cdot 29 \cdot 173$	173
197	78012	$3^2 \cdot 11 \cdot 197$	197
277	154012	$139 \cdot 277$	277
389	303420	$3 \cdot 5 \cdot 13 \cdot 389$	389
397	316012	$199 \cdot 397$	397
409	335380	$5 \cdot 41 \cdot 409$	409
457	418612	$229 \cdot 457$	457
509	519180	$3 \cdot 5 \cdot 17 \cdot 509$	509
521	543924	$3^2 \cdot 29 \cdot 521$	521
593	704484	$3^3 \cdot 11 \cdot 593$	593
613	752764	$307 \cdot 613$	613
617	762612	$3 \cdot 103 \cdot 617$	103, 617
641	823044	$3 \cdot 107 \cdot 641$	107, 641
701	984204	$3^3 \cdot 13 \cdot 701$	701
733	1076044	$367 \cdot 733$	733
757	1147612	$379 \cdot 757$	757
797	1272012	$3 \cdot 7 \cdot 19 \cdot 797$	797
821	1349724	$3 \cdot 137 \cdot 821$	137, 821
829	1376140	$5 \cdot 83 \cdot 829$	83
853	1456924	$7 \cdot 61 \cdot 853$	853
937	1757812	$7 \cdot 67 \cdot 937$	67
953	1818324	$3^2 \cdot 53 \cdot 953$	953
1009	2038180	$5 \cdot 101 \cdot 1009$	101
1021	2086924	$7 \cdot 73 \cdot 1021$	73, 1021
1033	2136244	$11 \cdot 47 \cdot 1033$	1033
1097	2409012	$3^2 \cdot 61 \cdot 1097$	1097
1109	2461980	$3 \cdot 5 \cdot 37 \cdot 1109$	1109
1117	2497612	$13 \cdot 43 \cdot 1117$	1117
1129	2551540	$5 \cdot 113 \cdot 1129$	113, 1129
1153	2661124	$577 \cdot 1153$	1153
1181	2791884	$3 \cdot 197 \cdot 1181$	197, 1181
1213	2945164	$607 \cdot 1213$	1213
1217	2964612	$3 \cdot 7 \cdot 29 \cdot 1217$	1217
1249	3122500	$5^4 \cdot 1249$	1249
1289	3325620	$3 \cdot 5 \cdot 43 \cdot 1289$	1289
1321	3492724	$661 \cdot 1321$	1321
1361	3707364	$3 \cdot 227 \cdot 1361$	227, 1361
1373	3773004	$3 \cdot 229 \cdot 1373$	1373
1381	3817084	$691 \cdot 1381$	1381
1453	4225324	$727 \cdot 1453$	1453
1481	4389684	$3 \cdot 13 \cdot 19 \cdot 1481$	1481
1553	4826724	$3 \cdot 7 \cdot 37 \cdot 1553$	1553
1597	5104012	$17 \cdot 47 \cdot 1597$	1597

q	N	odd part m	obstr.
1601	5129604	$3^2 \cdot 89 \cdot 1601$	89, 1601
1613	5206764	$3 \cdot 269 \cdot 1613$	269, 1613
1637	5362812	$3^2 \cdot 7 \cdot 13 \cdot 1637$	1637
1709	5844780	$3^2 \cdot 5 \cdot 19 \cdot 1709$	1709
1721	5927124	$3 \cdot 7 \cdot 41 \cdot 1721$	1721
1733	6010044	$3 \cdot 17^2 \cdot 1733$	1733
1741	6065644	$13 \cdot 67 \cdot 1741$	67
1753	6149524	$877 \cdot 1753$	1753
1777	6319012	$7 \cdot 127 \cdot 1777$	127, 1777
1789	6404620	$5 \cdot 179 \cdot 1789$	179, 1789
1873	7020004	$937 \cdot 1873$	1873
1877	7050012	$3 \cdot 313 \cdot 1877$	1877
1889	7140420	$3^3 \cdot 5 \cdot 7 \cdot 1889$	1889
1933	7476844	$967 \cdot 1933$	1933
1997	7980012	$3^3 \cdot 37 \cdot 1997$	1997
2017	8140612	$1009 \cdot 2017$	2017
2053	8433724	$13 \cdot 79 \cdot 2053$	2053
2081	8665284	$3 \cdot 347 \cdot 2081$	347, 2081
2113	8933764	$7 \cdot 151 \cdot 2113$	151, 2113
2129	9069540	$3 \cdot 5 \cdot 71 \cdot 2129$	71, 2129
2161	9344164	$23 \cdot 47 \cdot 2161$	2161
2213	9799164	$3^3 \cdot 41 \cdot 2213$	2213
2269	10301260	$5 \cdot 227 \cdot 2269$	227, 2269
2273	10337604	$3 \cdot 379 \cdot 2273$	2273
2281	10410484	$7 \cdot 163 \cdot 2281$	163
2293	10520284	$31 \cdot 37 \cdot 2293$	2293
2297	10557012	$3 \cdot 383 \cdot 2297$	383, 2297
2309	10667580	$3 \cdot 5 \cdot 7 \cdot 11 \cdot 2309$	2309
2341	10965244	$1171 \cdot 2341$	2341
2357	11115612	$3^2 \cdot 131 \cdot 2357$	131, 2357
2377	11305012	$29 \cdot 41 \cdot 2377$	2377
2381	11343084	$3 \cdot 397 \cdot 2381$	397, 2381
2389	11419420	$5 \cdot 239 \cdot 2389$	239, 2389
2417	11688612	$3 \cdot 13 \cdot 31 \cdot 2417$	2417
2437	11882812	$23 \cdot 53 \cdot 2437$	2437
2441	11921844	$3 \cdot 11 \cdot 37 \cdot 2441$	2441
2473	12236404	$1237 \cdot 2473$	2473
2477	12276012	$3 \cdot 7 \cdot 59 \cdot 2477$	2477
2609	13618980	$3^2 \cdot 5 \cdot 29 \cdot 2609$	2609
2621	13744524	$3 \cdot 19 \cdot 23 \cdot 2621$	2621
2657	14124612	$3 \cdot 443 \cdot 2657$	443, 2657
2677	14338012	$13 \cdot 103 \cdot 2677$	103, 2677
2689	14466820	$5 \cdot 269 \cdot 2689$	269, 2689
2693	14509884	$3 \cdot 449 \cdot 2693$	449, 2693
2713	14726164	$23 \cdot 59 \cdot 2713$	2713
2749	15119500	$5^3 \cdot 11 \cdot 2749$	2749
2777	15429012	$3 \cdot 463 \cdot 2777$	463, 2777
2789	15562620	$3^2 \cdot 5 \cdot 31 \cdot 2789$	2789
2797	15652012	$1399 \cdot 2797$	2797
2801	15696804	$3 \cdot 467 \cdot 2801$	467, 2801
2833	16057444	$13 \cdot 109 \cdot 2833$	109, 2833
2837	16102812	$3 \cdot 11 \cdot 43 \cdot 2837$	2837
2857	16330612	$1429 \cdot 2857$	2857
2909	16930380	$3 \cdot 5 \cdot 97 \cdot 2909$	2909
2917	17023612	$1459 \cdot 2917$	2917

None of the orders in Table 1 is a known open Hadamard order; by (4.1) each odd part is composite, while the open orders below 10^4 are $4 \times$ prime. Widening the ingredient set (computer-tabulated Williamson and T-matrices, or Goethals–Seidel cyclotomic difference sets) removes candidates from the table but can never produce one that meets the open list. The role of Theorem 1 is thus to give a single closed-form rule for the whole family, including the region these tables do not reach, rather than to settle any open existence question.

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